

Alice is a kindergarten teacher. She wants to give some candies to the children in her class. All the children sit in a line and each of them has a rating score according to his or her performance in the class.

Alice wants to give at least 1 candy to each child. If two children sit next to each other, then the one with the higher rating must get more candies. Alice wants to minimize the total number of candies she must buy.

Example

`arr = [4, 6, 4, 5, 6, 2]`

She gives the students candy in the following minimal amounts: `[1, 2, 1, 2, 3, 1]`. She must buy a minimum of 10 candies.

Function Description

Complete the `candies` function in the editor below.

`candies` has the following parameter(s):

- `int n`: the number of children in the class
- `int arr[n]`: the ratings of each student

Returns

- `int`: the minimum number of candies Alice must buy

Input Format

The first line contains an integer, `n`, the size of `arr`.

Each of the next `n` lines contains an integer `arr[i]` indicating the rating of the student at position `i`.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq arr[i] \leq 10^5$

Sample Input 0

```
3
1 2 2
2
2 1
```

Sample Output 0

```
4
```

kalau ratingnya sama? jumlah permeny-
harus sama, atau boleh beda? → kata
e.g 1, boleh beda.

? harus sendiri
urutan duduk
anak?

→ ga. udh dari

input aja, ga wrd
urutan 'low'



2 2 2 ?

= 1 1 1 ?

= 3 ?

Explanation 0

Here 1, 2, 2 is the rating. Note that when two children have equal rating, they are allowed to have different number of candies. Hence optimal distribution will be 1, 2, 1.

Sample Input 1

kalan ratingnya sama, boleh jd dpt jumlah permen yg sama

10
2
4
2
6
1
7
8
9
2
1

Sample Output 1

19

Explanation 1

Optimal distribution will be 1, 2, 1, 2, 1, 2, 3, 4, 2, 1

Sample Input 2

bln 1
2 km stlhny - maki ada yg ratingnya lebih rendah -

8
2
4
3
5
2
6
4
5

Sample Output 2

12

Explanation 2

Optimal distribution will be 12121212.

Suppose :

```
arr = [4, 3, 2, 1]
```

Then :

4 3 2 1

$\sqrt{\quad}$

2? perhatikan
 $\text{ad}_L(3)$ angle γ

3? to suppose = n

✓ (4?) → how do we know?

4 3 2 1

begitupula ini:

$$arr = [1, 2, 2]$$

1 2 2

17

✓ 2?

? gmn kalau case y
y stlhnya raty?
lebih kecil?

atau sama dengan

setelah 4, & turunnya
scr berhenti?

$$k \text{ m:}$$
$$arr = [4, \overline{3, 3, 2}]$$

4 3 3 2

A2

4 3? 2?

13

Kalau ini?: $arr = [4, 3, 3, 4]$

4 3 3 4

$3?$ $2?$ $1?$ 2 None
 $n.k.a$
 $n=0$
 $(1) + (1)$

if $arr[i] < arr[i+1]$:

~~$num_candy[i] = 1$~~ $\rightarrow$ udh pasti 1?

ga jg. tergantung rating anak
sebelumnya,

elif $arr[i] > arr[i+1]$ (?)

2 < 3, ^{contoh:} ~~typo~~ $\neq 1$

1 (2) 3

1 (2) 3

= sblm + 1 jk lebih tinggi
dari sblm

Suppose

1 4 3 (2) 2

1 2 (1?)
0

$num_lower_after = 2$

slm uk 2

ga, km ada 2 di sbh 3.

2? ga

(3) ya.

= $1 + num_lower_after$ = $1 + 2 = 3$

if $arr[i] > arr[i+1]$

~~if $arr[i+1] > arr[i]$~~

a.k.a

~~if $arr[i] < arr[i+1]$~~

ingat sbh

Tapi kalau gini gmn?

arr = [1 4 3 2 2 1]

ga berlaku lagi dong sbelum + 1 ^{anahd} / ratiy 4?

1 4 3 2 2 1
1 (2?)
1+1 =
num_lower_after = 4

1+4
hm.

if arr[i+1] > arr[i]:

if num_lower_after == 0:

num_candy[i+1] = sbelum + 1

elif num_lower_after != 0:

num_candy = 1 + num_lower_after

emang iya
hrs selalu "sbelum"?
kalau 0+1?

emang iya boleh selalu = 1?

~~kebay iya simpn
dulu, iya.~~

ga boleh selalu 1. cek pg 7

Konstruktion ulang:

```
if arr[i+1] > arr[i]:
```

```
    num_lower_after = 0
```

```
    j = i + 1
```

```
    for j in range n:
```

```
        if arr[j+1] < arr[j]:
```

```
            num_lower_after = + 1
```

```
        else:
```

```
            pass (?)
```

```
if num_lower_after == 0:
```

```
    num_candy[i+1] = sb/m + 1
```

```
elif num_lower_after != 0:
```

```
    num_candy = 1 + num_lower_after
```

```
elif arr[i+1] <= arr[i]:
```

3 1 2 ?
 ?
 → ps next
 (1) ✓

3 2 (1)

3 2 1

~~(1) ?~~

elif $arr[i+1] \leq arr[i]$: ?

5 ② ① 3
③ ② 1 2

num_higher_after = 3

1 2 3 4
0+1=1= ① ② 3 4
 └ 1+1= └ 2+1= └ 3+1=

2 3 4 4

1 2 ③ 1

└ gaboleh

= ~~X~~+1, soalnya ada 2 candies sblmnya

& rating yg diberikan 2 candies

itu lebih kecil dari anch rating 4,

jadi candies nya hrs > 2.

a.k.a

= ~~sbm + num_lower_after~~

Tapi kalau $\text{num-lower-after} > \text{jumlah candies dan rating}$

e.g.

2 3 4 (4 3)

1 2 3

||

$1 + \text{num-lower-after}$

$= 1 + 2 = 3$ 0-0

yg lebih kecil & sebelum

anak rating 4, bknny.

Jdny - akhirnya = sah?

aja pake $= 1 +$

$\text{num-lower-after}?$

if

gmn dgn case $\text{num-lower-after} = 0?$

$3 > 1$

1 [3]

Vs

$3 < 4$

4 [3]

1 (2)

2 (1)

$\therefore \text{sbkm} + 1$

$\neq \text{sbkm} + 1,$
(ainkan

$= 1$

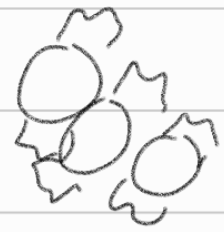
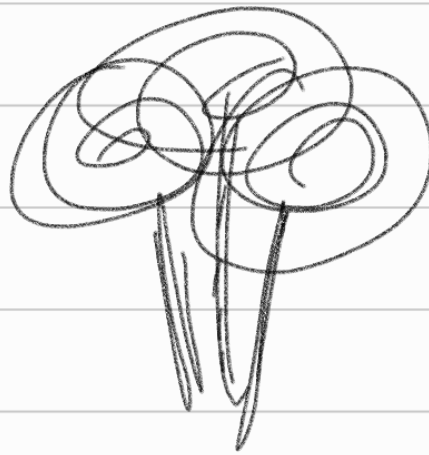
$3 > 2$

1 2 3

1 2 (3)

$= \text{sbkm} + 1$ ✓

coba ulay lagi.



$$1 < 6 \& 1 < 7$$

$$2 < 9 \& 2 > 1$$

2	4	2	6	(1)	7	(8)	(9)	(2)	1
1	2	1	2	1	2	3	4	2	1

$$8 > 7 \& 8 < 9$$

$$9 > 8 \& 9 > 2$$

eh tapi kalau gini? :

2	4	2	6	1	7	8	(9)	5	4	3	2	1
1	2	1	2	1	2	3	(4)					

brarti sedari
anakini hrs
udh mulai

'diperhatikan'.

ga boleh = 4 krn $9 > 5$

ga boleh = 3 krn anak? stlhmy =

ga kedapetan permen D:

(malah minus!)

ngutag permen wk

pakai rule

mm-lower-after

Jika ga boleh opt + anak = 4 permen - stlhmy.

num_candies = []

3 2 1

for i in range(n):

for i in range 2

if i == 0:

0, 1, 2

if arr[i] ≤ arr[i+1]:

0 ≤ 1

0 ≤ 2

num_candies.append(1)

elif arr[i] > arr[i+1]:

num_candies.append(1 + num_lower_after[i])

elif i == n-1:

if arr[i] ≤ arr[i-1]:

num_candies.append(1)

elif arr[i] > arr[i-1]:

num_candies.append(num_candies[i-1] + 1)

else:

if ~~arr[i] > arr[i-1]~~ and arr[i] > arr[i+1]:

num_candies.append(1 + num_lower_after[i])

elif arr[i] ≤ arr[i+1]:

num_candies.append(num_candies[i-1] + 1)

$$arr = [1, 2, 2]$$

0	1	2
---	---	---

$$\boxed{n=3}$$

for i in range(3)

$$i=0 \rightarrow arr[0] \leq arr[1]$$
$$i = 1 \rightarrow \text{arr}[i] \leq \text{arr}[v]$$

$i = 2 \rightarrow arr[2] \leq arr[3] ??$

$$i = n?$$

arr	2	4	2	6	1	7	8	9	2	1
num_lower	0	1	0	1	0	0	0	2	1	0
candies	1	2	1	2	1	2	3	4	2	1

3+1

How to determine num_lower_after? 

num_lower_after = []

counter = 0

for i in range(n ~~+~~):

range
0 s.d n-1
3

~~j = i~~

for j in range(i, n-1 ~~> j~~):

range
3 s.d n-1-i

if arr[j] > arr[j+1]:

counter += 1

i=0 j=1 j=2 i=3
4 2 3 1

elif arr[j] == arr[j+1]

j=0

counter += 0

else:

~~pass~~ break -,,- astaga

num_lower_after.append(counter)

(walaun
bisa)

counter = 0

tapraastagany < lusi, ini ga

lolos test 11. Time limit

test : arr : [4 3 2 1 2 3]

exceeded.

3 2 1 0 0 0

solving?

see pg 15

$n = 8$

arr =

	0	1	2	3	4	5	6	7
	2	4	3	5	2	6	4	5

num_lower_after =

	0	1	0	1	0	1	0	0
--	---	---	---	---	---	---	---	---

candles

	1	2	1	2	1	2	1	2
--	---	---	---	---	---	---	---	---

$\text{arr}[2] > \text{arr}[3] ? \rightarrow \text{no}$

$\text{arr}[2] < \text{arr}[3] ? \rightarrow \text{yes}$

$i=0, j=0 \rightarrow \text{arr}[0] > \text{arr}[1]$

7 8-1

for j in range(1, 8-1) 5,

$i=5 \rightarrow j=5$ 8-1 7

$j=5 \rightarrow \text{arr}[5] > \text{arr}[6] ?$

$j=7 \rightarrow \text{arr}[7] > \text{arr}[8] ?$

if $j == n-1$:

pass

0 1 2

1 2 2

1 2 1

$\text{arr}[1] > \text{arr}[2] ? \text{yes}$

akhirnya = nanya gpt krn udh mentok bgt/lha.

o oke lah -
isoke.

(& bagian nyari
(num_lower_after)

yg penting ngerti & implementasi

& masa depan nanti kalau nemu

lagi case gitu



```
num_lower_after = [0] * n
```

```
for i in range(n-2, -1, -1):
```

```
    if arr[i] > arr[i+1]:
```

```
        num_lower_after[i] = num_lower_after[i+1] + 1
```

katanya ini trik dynamic programming

suppose $n=5$, $arr = [5, 4, 3, 2, 1]$

bruti :

$num_lower_after = [0]^* 5 = [0, 0, 0, 0, 0]$

for i in range($5-2, -1, -1$) :

range(3, -1, -1)

$= 3, 2, 1, 0$ jalannya dari kanan ke kiri,
blm yg kekekrasaja 0, 1, 2, 3

$i=3 \rightarrow arr[3] > arr[4] ?$ ya

$\rightarrow num_lower_after[3] = num_lower_after[4] + 1$
 $= 0 + 1 = 1$

$i=2 \rightarrow arr[2] > arr[3] ?$ ya

$\rightarrow num_lower_after[2] = num_lower_after[3] + 1$
 $= 1 + 1 = 2$

$i=1 \rightarrow$ 1 2 ya

$= 2 + 1 = 3$

$i=0 \rightarrow$ 0 1 ya

$= 3 + 1 = 4$

$\therefore num_lower_after = [4, 3, 2, 1, 0]$